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A Typological Perspective on Standard Negation in an Emerging Sign Language

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1 Introduction

All languages, both spoken and signed, have negation as a universal feature (Horn 2001). Negation constitutes a key grammatical domain since speakers of any language need to negate sentences (Zeshan 2006). Thus, negative structures have been extensively studied across spoken and signed languages, making typological comparison possible. The substantial literature on sign language negation addresses different aspects such as the types of markers (i.e., manual vs. nonmanual) and syntactic patterns. Manual markers refer to use of hands while nonmanual markers (henceforth NMMs) are facial expressions and body movements (Zeshan 2006). NMMs serve various phonological, morphological, syntactic, and prosodic functions (Pfau 2016, Wilbur 2021). Sentential negation uses two components: manual markers and nonmanual markers (e.g., Bembridge 2016, Pfau 2016, Pfau *et al.* 2022, Quer 2012). Manual markers indicate negative meanings, ranging from basic negative operators to specific ones. The nonmanual negator is mostly a headshake which can spread over the manual negator only or further (over the VP or the entire clause). This spread/scope is defined as “the extent of a string of manual signs co-occurring with the non-manual marker” (Zeshan 2006:39).

Research has shown that there is extensive variation in how manual and nonmanual markers interact. Sign languages appear divided into two types: manual-dominant and nonmanual-dominant (Zeshan 2006:43; Table 1). In manual-dominant sign languages, a manual negator is required to negate a sentence, although a NMM might accompany it. Italian Sign Language (LIS), Jordanian Sign Language (LIU), Turkish Sign Language (Türk İşaret Dili, TİD), Kata Kolok (Bali), and Japanese Sign Language (JSL) fall into this category. In nonmanual-dominant sign languages, negation might use both manual and nonmanual markers but is possible through the nonmanual negator alone, as in American Sign Language (ASL), German Sign Language (DGS), Catalan Sign Language (LSC), and Swedish Sign Language (Svenska Teckenspråket), amongst others.

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Nonmanual dominant negation systems	Manual dominant negation systems
nonmanual negation is obligatory	nonmanual negation is not obligatory
clause can be negated nonmanually only, manual basic clause negator is optional	clause cannot be negated nonmanually only, manual negator is required
choice of nonmanual marking does not depend on manual signs	choice of nonmanual marking depends on choice of manual clause negator (if there is more than one nonmanual configuration)
nonmanual negation spreads freely over the clause	scope of nonmanual negation is over the manual negator only or is closely tied to the manual negator
Examples: DGS, ASL, Australian Sign Language (Auslan)	Examples: JSL, Kata Kolok, TİD

Table 1: Manual dominant and nonmanual dominant systems of negation

Furthermore, there are differences with respect to (a) the morphological nature of the negative marker and (b) position with regard to the verb (Pfau 2016). For instance, negation is postverbal in DGS and TİD, but preverbal in LIS and Berber Sign Language. However, in general, negative signs tend to occur sentence-finally (Zeshan 2006) and are usually accompanied by such nonmanual activities as frowning; squinted eyes; nose-wrinkling; lips spread, pursed or with the corners down; puffed cheeks; air puff; tongue protruding; etc. (Quer 2012). For example, in American Sign Language (ASL), the basic syntactic negator, ‘NOT’, can occupy either a preverbal position or a sentence-final position if the rest of the sentence is topicalized and is almost always accompanied by a headshake (hs) (Fischer 2006), but also frown, brow squint, wrinkling of the nose, and a raised upper lip (Aarons 1994). Thus, the negative headshake may be used along with the lexical negator NOT. In such cases, the headshake does not have to spread over the entire VP as seen in (1):

- (1)

hs

JOHN NOT BUY HOUSE

‘John did not buy a house.’
- (ASL)

(Pfau & Quer 2002:75)

However, in the absence of the manual negator NOT, the headshake can negate the sentence (Fischer 2006). In such cases, the headshake must extend over the entire verbal domain:

- (2)

hs

JOHN BUY HOUSE
- (ASL)

(Pfau & Quer 2002:75)

Pfau & Quer (2002) argue that the verb *never* raises to Neg⁰ in ASL. Rather, in the presence of NOT, the realization of [+neg], i.e., headshake, associates with NOT and the sentence is grammatical. However, the absence of NOT would mean that there is no manual material that [+neg] can be articulated with and therefore the headshake would be forced to spread over its entire c-command domain, as represented schematically in Figure 1.

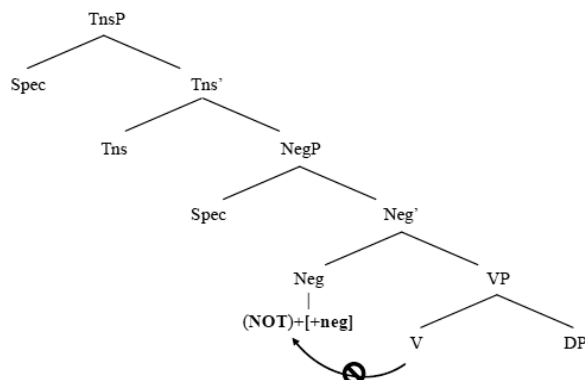


Figure 1: Negation in ASL (Pfau & Quer 2002:77)

This analysis explains why (3) is ungrammatical in ASL: NOT is absent, *hs* has no lexical material to associate with, yet it has not spread over its c-command domain:

- (3) *hs*
*JOHN BUY HOUSE (adapted from Pfau & Quer 2002:77)

The above observations show that ASL is characterized by all the features of a nonmanual dominant negation system.

1.2 Context and objective of study

In this study, we sketch the realization of sentential negation in a young sign language. Here, sentential negation refers to negation of simple non-embedded active predicative declarative sentences (Dahl 1979). Other types fall outside the current focus. This young sign language started to emerge about sixty years ago in the small village *Sadat Tawaher* in the province of Khuzestan in southwestern Iran and is referred to as *Sadat Tawaher Sign Language (STSL)*. Our objective is to show that STSL is manual-dominant based on different types of evidence. *Sadat Tawaher* has a population of about 400 people and the only spoken language is Khuzestani Arabic (KhA). STSL started to develop after an adult man in his early 20s named *Hanash* lost his hearing, making STSL about 55+ years old. Hanash lived to be¹ 76 years old and had nine children. Our estimate of STSL's age is based on the fact

¹ Sadly, Hanash passed away on December 27, 2021, at age 76.

that none of the children, especially his oldest child who is now 54, remembers when he lost his hearing, and therefore he must have become deaf at least 55 years ago. Currently, more than 30 people are proficient in STSL. Hanash spoke KhA natively but had no literacy and had been widowed for 30 years. Since all other family members are hearing, they needed to sign to him. However, since he could speak, he did not have to sign to them. Thus, the signing is one-way. Hanash was the only deaf person in Sadat Tawaher.

2.2 KhA and STSL: Brief introduction to grammar

KhA is a pro-drop dialect of Arabic spoken in the Iranian province of Khuzestan. Like other Arabic varieties, KhA has a flexible word order, allowing SVO and VOS ((4a) and (4b)), respectively), and to a lesser extent, VSO. SOV is also allowed.

- (4) a. SVO: il-walad kala t-tiffaahah
 the-boy ate.3MS the-apple
 'The boy ate the apple.'
 b. VOS: kala t-tiffaahah il-walad
 ate.3MS the-apple the-boy

In contrast to KhA, STSL has two common word orders: SVO and SOV. SVO is used far more frequently than SOV and with all types of verbs and is thus likely the basic word order. SOV is especially used with morphologically marked agreeing verbs, classifier constructions, and locatives. OSV order with O topicalized is also allowed with the verb mostly showing agreement with the object through eye gaze, indexing, etc.

STSL seems to be a head-final language. Evidence comes from the fact that elements such as the *wh*-question marker, perfective aspect marker *FINISH*, and negation markers appear in the right periphery in sentence-final position and determiners follow their complements. Dropping overt pronominal subjects (and even objects) is very common. Modals mostly appear in preverbal position (i.e., S-Mod-VO or S-Mod-OV) (5).

- (5) a. SVO: BOY EAT APPLE
 b. SOV: BOY APPLE EAT
 'The boy ate the apple.'
 c. OSV: APPLE BOY EAT
 'The apple, the boy ate.'

2.3 Research question and hypothesis

Our overarching hypothesis is that negation in STSL is realized similarly to negation in other sign languages because STSL is *not* a copy of the spoken language, KhA; that is, it is not *Signed KhA*, rather an independent language. We formulate the following research question: How does negation in STSL compare to

that in other sign languages?, i.e., where does STSL fit in the dichotomy of manual dominant vs. nonmanual dominant sign languages?

We claim that STSL is typologically manual-dominant. We show that the manual negator is obligatory and able to negate the sentence on its own while the nonmanual negator is optional and unable to negate the sentence on its own or to spread over adjacent signs or constituents.

3 Methodology

3.1 Participants

Five STSL signers including 3 males and 2 females (age range: 27-54) participated. Except for one signer, all the participants were native signers of STSL, having signed it since birth. The one non-native signer is completely fluent in STSL and has used it for 30 years since she married Hanash's oldest son. All the participants speak KhA as their mother tongue and all but one live in Sadat Tawaher. Three participants have no education, one has a bachelor's degree in English, and one elementary education. As for other languages, three participants only speak KhA, one participant speaks Farsi fluently, and one is fluent in Farsi and English. As for their relation to Hanash, one is his son, one his daughter, two his grandsons, and one his daughter-in-law. All the participants had constant interaction with Hanash and only used STSL to communicate with him.

3.2 Data

We used multiple methods to collect various types of data as a strategy to enhance the validity of the results. Thus, we gathered three types of data: (i) signed productions, (ii) story-telling, and (iii) grammaticality judgments.

3.2.1 Signed productions

The signed productions were a set of 36 stimulus sentences constructed in KhA and shared with four participants via WhatsApp. The stimuli were submitted in written form to one participant who could read and in audio form to the three others as they did not have reading literacy. Included were simple declarative active voice sentences with either (di)transitive or intransitive verbs, where transitive verbs consisted of unmarked verbs (e.g., drink, love), marked agreeing verbs (e.g., push, give), body-anchored locatives (e.g., poke, slap), and classifier constructions (e.g., put), while the intransitive verbs included 'sleep' and 'fall' and locatives (e.g., next to, on top of). The sentences consisted of reversible (e.g., 'The man didn't see the doctor') and non-reversible predicates (e.g., 'My friend doesn't drink tea'). In total, we received 139 signed sentences, of which 124 were useable for analysis.

3.2.2 Story-telling

Story-telling was to ensure that signers would not be restricted by fixed sentences but would have total freedom to express what they wished to narrate in more natural

productions. The participants were asked to produce as many stories as possible of two types: Type I: bedtime stories and Type II: everyday activities. Bedtime stories ('Stories of the Levant') are meant to be amusing but also communicate or teach a moral principle. Overall, we obtained 74 videos of bedtime stories from four participants (32 unique stories, total length 01:17:38) with 148 negative simple sentences in them. For everyday activities, the participants were asked to narrate an everyday activity, something that takes place on a daily basis, or has happened in the recent or far past, for instance, wheat harvest procedure, a recent car accident, etc. We gathered 283 everyday activity videos from four participants comprising 189 unique activities/incidents totaling 03:32:23 in length. The participants produced 777 negative simple sentences in these stories. In sum, we obtained 357 videos consisting of 221 stories/activities, totaling 04:50:01, with 925 negative sentences.

3.2.3 Grammaticality judgments

Using grammaticality judgments was a way to make sure that what the participants considered ungrammatical in STSL would be clearly and directly stated. We collected these judgments from two participants, one male and one female, one directly via Google Meet and the other by submitting videos via WhatsApp for judgments.

4 Results and discussion

STSL has multiple manual negation markers, discussed in Section 4.1. Section 4.2 discusses the manual and nonmanual negators in STSL's standard negation.

4.1 Negation markers

STSL has six different negators: NEG_{basic} , NEG_{ref} , NEG_{proh} , NEG_{emph} , NEG_{poss} , and NEG_{exist} . For each negator, we have incorporated the *main* pragmatic function as a subscript although some negators have more than one pragmatic function. Each negator consists of a manual element and one (or more) nonmanual element(s) associated with the manual element. We demonstrate that all negators appear in sentence-final position.

We argue that NEG_{basic} is the basic clause negator in STSL for two reasons: (a) it is used to change the polarity of a pragmatically neutral sentence without adding any pragmatic or semantic flavor to the sentence; (b) it can negate sentences with a wide range of semantic functions (existential, possession, etc.). Configurationally, it consists of a single upward movement of both hands or only the dominant hand from the wrist and a backward head-tilt (bht) simultaneously.

Fig. 2: NEG_{basic}

- (6) $\text{IX-1 SMOKE } \overline{\text{NEG}_{\text{basic}}}^{\text{bht}}$
 ‘I do not smoke.’

NEG_{ref} consists of a single rightward movement of the dominant hand and a single leftward head turn (ht) (i.e., in opposite directions) and expresses refusal to do an action. It might also be accompanied by pouted lips and furrowed brows.

- (7) $\text{CHILD EAT BREAKFAST } \overline{\text{NEG}_{\text{ref}}}^{\text{ht}}$
 ‘My child refuses to eat breakfast.’

NEG_{proh} consists of repeated side-to-side movement of both hands or only the dominant hand and a simultaneous side-to-side headshake (hs) and expresses prohibition, warning, or simply emphatic negation.

- (8) $\text{MEAT FRIEND EAT } \overline{\text{NEG}_{\text{proh}}}^{\text{hs}}$
 ‘My friend never eats meat.’

NEG_{emph} is similar to NEG_{proh} in its pragmatic function and movement of articulators except for the handshape, which is composed of the repeated, fast wagging of the index finger of the dominant hand and brow raise. It does not appear to serve for clausal negation.

- (9) a. MONEY IX-2 STEAL? ‘Did you steal the money?’
 b. NEG_{emph} ‘No!’

NEG_{poss} consists of the tips of all fingers of the dominant hand swiping down the tip of the nose with air blown out (ab) at the same time. Functionally, it is used to indicate lack of possession (‘not-have’) and non-existence (‘there is/are not’).

- (10) $\text{MAN IX-3 } \overline{\text{NEG}_{\text{poss}}}^{\text{ab}}$
 ‘That man has nothing (i.e., is very poor).’

NEG_{exist} uses the completive aspect marker *FINISH* as a negative existential marker and suggests primarily non-existence but also lack of possession. NEG_{exist} is interchangeable with NEG_{basic} but seems to have a different pragmatic function: NEG_{exist} implies a sense of unexpectedness while NEG_{basic} simply negates the existence of something, supporting NEG_{basic} as the basic negator. NEG_{exist} is signed with both hands swiping each other up and down and might be accompanied by headshake.

- hs

- (11) CHILD YARD NEG_{exist}
‘There are no kids in the yard.’

4.2 Standard negation

In this section, we present the findings regarding standard negation in STSL from each data collection procedure.

4.2.1 Signed productions data

Although STSL has six negation markers, the participants only used NEG_{basic} to negate the sentences in the signed productions task (Table 2). This finding is not surprising because the sentences in this task were pragmatically neutral statements (e.g., ‘The boy didn’t give the book to the girl’, ‘The man didn’t die’, ‘It is not raining’, etc.) This can be taken as evidence that this negator is the basic negator (further evidence to be presented later).

Negation markers				Word order & position of NEG						Other NMMs	
Manual marker: NEG _{basic}		Nonmanual marker (backward head-tilt)		SVO-Neg	SOV-Neg	SV-Neg	SV-Neg-O	S-Neg-V/O-Neg	SVOV-Neg	Brow raise	Eye gaze
Overt	Non-overt	Overt	Non-overt								
105	8	26	104	42	48	23	2	1	1	52	5

Table 2: Sentential negation from signed productions

Our findings show that NEG_{basic} is occasionally accompanied by its associated NMM, i.e., backward head-tilt (bht). Overall, the manual negator is overtly used in 105 sentences but it is non-overt in 8 sentences. The nonmanual negator is overt in only 26 sentences and non-overt in 104 sentences. These results provide support for our hypothesis that STSL is manual-dominant as the vast majority of the negative sentences have an overt manual negator while the nonmanual negator is either mostly absent or, if present, is very slightly visible and scopes over the manual negator only. Note that the primary condition that needed to be met for

STSL negation system to be considered manual dominant was the presence of manually negated sentences with no nonmanual negator present. We did not have a single sentence that was negated by a nonmanual negator only. Sentences with a non-overt manual negator did not have a nonmanual negator either. The following are examples of negative sentences from the signed productions.

(12)



PEOPLE

HARVEST

RICE

NEG_{basic}

'The people did not harvest the rice.'

(13)



SOLDIER

MAN

SHOOT

NEG_{basic}

'The soldier did not shoot the man.'

The 8 sentences with a non-overt manual negator could be participant mistakes due to distraction or some other factor, such as the fact that they almost always speak KhA when they sign. Thus, they might have erroneously thought they did produce the manual negator as they heard themselves say the negative word. It could also be that they assumed the discourse provided enough context for the deaf person to interpret the clause as negative. We will provide evidence for this claim below from the story-telling data and grammaticality judgments. In addition to bht, brow raise, and, less frequently, eye gaze were also seen with negation. The negation marker appeared sentence-finally in almost all the sentences.

4.2.2 Story-telling data

Altogether, the participants produced 925 negative sentences (Table 3). Of these, 680 (73.5%) use NEG_{basic} while the remaining five negators combined (plus the category 'other' which did not fall into any of the other five negators) constitute only 245 (26.5%) of all the negative sentences from the stories. Thus, the very high

frequency of usage of NEG_{basic} should be taken as further evidence for it being the basic clause negator in STSL.

	Total	NEG _{basic} c	NEG _{ref}	NEG _{proh}	NEG _{emph}	NEG _{poss}	NEG _{exist}	Other	No overt Neg
Stories I	148	93	0	2	0	10	16	27	22
Stories II	777	587	0	31	0	47	27	85	50
Stories I + II	925	680	0	33	0	57	43	112	72

Table 3: Clausal Neg from stories

Once again, the primary negator in all negative sentences from the stories was the manual negator. The nonmanual negator never negated the sentence on its own. Furthermore, in the few cases that the nonmanual negator accompanied the manual negator, it was mostly slightly visible, as in the headshake in (14), and never spread beyond the manual negator.

(14)



TEACHER BEAT IX-1 SCARED VERY

‘The teacher punished me, and I was very scared.’



TODAY OLD REMEMBER FORGET NEG_{proh}

‘To this day, that I’m old now, I still remember it. I have not forgotten at all.’

The following is an example of a bedtime story which narrates the account of a man who takes his elderly father to a remote area to leave him there because he is tired of looking after his father. The moral of the story is karma, i.e., how one treats one’s parents is how one’s children will treat one:

- (15) SAY MAN FATHER CARRY TAKE GO LAND BIG HOUSE OTHER **NEG_{exist}**. HOUSE PEOPLE CAR **NEG_{exist}**. GROUND **NEG_{proh}**. TREE ONE BIG. FATHER PUT TREE NEXT-TO, LEAVE. GO LITTLE. FATHER LAUGH. SON LOOK-BACK. SAY LAUGH WHY. SAY LAUGH PAST ME FATHER PUT HERE. NOW YOU ME PUT HERE. FUTURE SON YOU PUT HERE. FEEL SORRY. COME CARRY GO. LEAVE TREE UNDER **NEG_{basic}**.

‘It is said that a man carried his father and took him to a big area where there were no houses around. There were no houses, people, or cars. There was nothing in that area. There was one big tree. The man put his father down next to the tree and left. He walked a little. The father laughed. The man looked back and asked why he was laughing. The father said, “I’m laughing because in the past I also left my father here. Now you left me here. In the future, your son will leave you here.” The man felt sorry. He went back [to his father], carried him back and left. He didn’t leave him under the tree.’

In the above story, the negation markers appear in sentence-final position unaccompanied by any associated NMMs. In cases where the nonmanual negator accompanied the manual negator, it never spread over any adjacent signs.

4.2.3 Grammaticality judgment data

For grammaticality judgments, we chose two negators, namely **NEG_{basic}** and **NEG_{proh}**, to test our hypothesis that STSL is manual-dominant. Our choice is justified on the following facts: (i) **NEG_{basic}** and **NEG_{proh}** are both very commonly used; (ii) they both have various semantic functions and can be used in almost any context (e.g., existence, possession); (iii) the NMMs associated with **NEG_{basic}** and **NEG_{proh}** (backward head-tilt and headshake, respectively) can commonly spread in many sign languages; and, (iv) **NEG_{poss}** and **NEG_{exist}** seem reserved for more specialized contexts (e.g., possession or existence) and their NMMs are hard to spread even physiologically (imagine blowing air over the entire VP/sentence for **NEG_{poss}**!). We tested manual vs. nonmanual dominance with the tests shown in Table 4.

Test	Result
(a) Sentence-final neg + hs: FRIEND EAT MEAT $\overline{\text{NEG}_{\text{proh}}}$ ^{hs}	Grammatical
(b) Sentence-final neg without hs: FRIEND EAT MEAT NEG_{proh}	Grammatical
(c) Sentence-final neg + spreading hs: FRIEND $\overline{\text{EAT MEAT NEG}}$ ^{hs}	Ungrammatical
(d) Only hs; no spread; no manual neg: FRIEND EAT MEAT ^{hs}	Ungrammatical
(e) Spreading hs; no manual neg: FRIEND $\overline{\text{EAT MEAT}}$ ^{hs}	Ungrammatical
(f) Sentence-initial neg with(out) hs: $\overline{\text{NEG}}$ FRIEND EAT MEAT ^(hs)	Ungrammatical
(g) Preverbal neg with(out) hs: FRIEND $\overline{\text{NEG EAT MEAT}}$ ^(hs)	Ungrammatical

Table 4: Results of grammaticality judgments for negation

To elicit metalinguistic data, we also asked participants direct questions about how negative sentences are signed. When asked how to sign a negative sentence, one participant said the following: “To say NO, use your hand or your hand and head, but not just your head.” When we allowed hs or bht to spread over the entire clause or VP, he reacted by saying: “Don’t keep shaking your head! Only shake your head when you sign NO with your hand”. We further asked where in the sentence NO should appear to which he responded: “Sign what you want to say, and then sign NO.” Finally, when we asked why he signed differently from how he spoke Arabic (because negation is preverbal in Arabic), he responded: “We don’t care about Arabic. Arabic is different!”

4.3 Discussion

We hypothesized that STSL is manual-dominant, which requires the following criteria to be met (Pfau 2016, Pfau *et al.* 2022, Zeshan 2006): (a) the manual negator is required to negate the sentence (16) while the nonmanual negator is not only optional but also often left out (17); (b) the manual negator can negate a clause on its own (17) while the nonmanual negator cannot (18b); and (c) the nonmanual negator never spreads beyond the manual negator (18a) (bht = backward head tilt):

- (16)

bht

BROTHER DRINK TEA $\overline{\text{NEG}_{\text{basic}}}$

‘My brother does not drink tea.’
- (17)

BROTHER DRINK TEA $\text{NEG}_{\text{basic}}$
- (18)

(a) *BROTHER $\overline{\text{DRINK TEA}}$ ^{bht}

(b) *BROTHER DRINK TEA^{bht}

Therefore, we conclude that STSL is a strictly manual-dominant sign language. Following Pfau (2016), we argue that since the nonmanual negator is optional, the obligatory manual marker occupies the Neg^0 position of NegP . On the other hand, the nonmanual negator, e.g., backward head-tilt, requires that lexical material be associated with it because it is affixal in nature; in fact, all of the nonmanual negators have an uninterpretable feature $[\text{uNEG}]$, which is why they cannot negate a sentence on their own. The NMMs probe for an interpretable feature of negation, which is found on the manual negator. By associating with the manual negator, the nonmanual negators check their uninterpretable feature through a Spec-Head Agree relation. In STSL, the head position of NegP may always be occupied by the manual negator (not void of phonological material). In this regard, STSL's negation system is similar to other sign languages such as TİD (Gökğöz 2011).

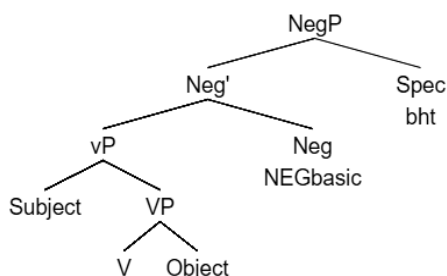


Fig. 3: Projection of negation in STSL

Furthermore, the positions occupied by each negator are also congruent with syntactic theory, i.e., manual negator occupying the head of NegP and nonmanual negator the specifier position, since heads are obligatory and specifiers optional.

5 Conclusion

The findings clearly suggest that STSL is a manual-dominant sign language, aligning with other sign languages such as TİD, LIS, and LIU. In STSL, negation is realized with an obligatory manual negator which is rarely accompanied by a nonmanual negator. STSL does, however, rely heavily on nonmanual negators alone, e.g., headshake, backward head-tilt, etc., as response particles in the absence of a manual negator but never for clausal negation. This might be evidence that STSL makes a distinction between negation at sentential and non-sentential levels. In line with arguments put forward for other sign languages (e.g., Auslan, Johnston 2018), the nonmanual negator does not carry negative force in STSL but the manual negator does. The nonmanual negators seem to be gestures with no particular syntactic function. In conclusion, our findings lend support to our claim that STSL and KhA are two *different* languages; that is, STSL is not *Signed KhA*, but a true sign language independent of KhA in terms of its grammar, in this case in terms of

negation. Even though STSL grew in the vicinity of KhA and was likely structurally based on KhA at least initially, it is grammatically different from KhA. One might attribute such grammatical differences to the fact that STSL and KhA use two different modalities, i.e., visual-spatial vs. auditory-vocal.

5.1 Future directions

The current research is the first study on STSL. There remain issues on negation that future research can address. For instance, one can examine whether the occurrence of two, or even more, manual negators in the same clause (i.e., negative concord (NC)) is allowed in STSL without affecting the polarity of the sentence; that is, with the sentence remaining negative. Since no sign language has so far been classified as a double negation (DN) language (Pfau 2016), we anticipate that STSL allows NC structures. If our prediction is correct, an important question to address in the future would be to examine what combinations of negation markers are allowed in STSL and for what semantic functions. Since, based on evidence from frequency of usage and semantic functions, we argued that NEG_{basic} is the basic clause negator in STSL, we hypothesize that in NC structures, if allowed, NEG_{basic} must always be the first negator, e.g., $NEG_{basic} + NEG_{proh}$. Thus, we predict the following:

- (19) BROTHER BREAD EAT $\frac{bht}{NEG_{basic}} \frac{hs}{NEG_{proh}}$
 'My brother doesn't eat bread at all.'
- (20) *BROTHER BREAD EAT $\frac{hs}{NEG_{proh}} \frac{bht}{NEG_{basic}}$

We further hypothesize that the second negator in NC sentences like NEG_{proh} and NEG_{ref} would have to have acquired pragmatically specialized meanings (warning, prohibition, refusal, denial, etc.) as also reported for other sign languages. In other words, NEG_{basic} simply changes the polarity of the sentence while the other negators, e.g., NEG_{proh} and NEG_{ref} , not only do so (when they do not serve as negative adverbials in NC constructions but as sentential negators), but also add more specialized readings if combined with NEG_{basic} . If STSL, like other sign languages, is also an NC language, then this would have implications for the NegP projection: If, for instance, NEG_{basic} and NEG_{proh} each headed their own NegP projection, a double negation (DN) reading would result (Aoun *et al.* 2010) which, we predict, is not allowed in STSL. We will address these questions in future research.

References

- Aarons, D. 1994. *Aspects of the Syntax of American Sign Language* (Doctoral dissertation, Boston University).
- Aoun, J., Benmamoun, E., & Choueiri, L. 2010. *The Syntax of Arabic*. Cambridge: Cambridge University Press.
- Bembridge, G. 2016. Negation in American Sign Language: The view from the interface. *Toronto Working Papers in Linguistics (TWPL)* 36, 1–20.
- Dahl, Ö. 1979. Typology of sentence negation. *Linguistics* 17, 79–106.
- Fischer, S. D. 2006. Questions and negation in American Sign Language. In *Interrogative and Negative Constructions in Sign Languages*, ed. by U. Zeshan, 165–197. Nijmegen: Ishara Press.
- Gökgöz, K. (2011). Negation in Turkish Sign Language: The syntax of nonmanual markers. *Sign Language & Linguistics* 14(1), 49–75.
- Horn, L. R. 2001. *A natural history of negation*. Chicago, IL: University of Chicago Press.
- Johnston, T. 2018. A corpus-based study of the role of headshaking in negation in Auslan (Australian Sign Language): Implications for signed language typology. *Linguistic Typology* 22, 185–231.
- Pfau, R. 2016. A featural approach to sign language negation. In *Negation and Polarity: Cognitive and Experimental Perspectives*, ed. by P. Larrivé & C. Lee, Vol. 1, 45–74. Dordrecht: Springer.
- Pfau, R., Makharoblidze T., & Zeijlstra, H. 2022. Negation and negative concord in Georgian Sign Language. *Frontiers in Psychology* 13, 1–18.
- Pfau, R., & Quer, J. 2002. V-to-Neg raising and negative concord in three sign languages. *Rivista di Grammatica Generativa* 27, 73–86.
- Quer, J. 2012. Negation. In *Sign Language: An International Handbook*, ed. by R. Pfau, M. Steinbach, & B. Woll, 316–339. Berlin: De Gruyter Mouton.
- Wilbur, R. B. 2021. Non-manual markers: Theoretical and experimental perspectives. In *Routledge Handbook of Theoretical and Experimental Sign Language Research*, ed. by J. Quer, R. Pfau, & A. Herrmann, 530–565. Abingdon: Routledge Taylor & Francis.
- Zeshan, U. 2006. Negative and interrogative constructions in sign languages: A case study in sign language typology. In *Interrogative and Negative Constructions in Sign Languages*, ed. by U. Zeshan, 28–68. Nijmegen: Ishara Press.

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Kutay Serova
M. K. Snigaroff

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- The Center for Leadership and Involvement
- The Franke Institute for the Humanities
- The University of Chicago Graduate Council

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We are very grateful for each conference presenter and attendee. This year, undergraduates, graduate students, and faculty alike attended, and their thoughtful comments, questions, and perspectives are what make CLS what it is each year. We are particularly grateful to those who traveled to join us, as we had in attendance scholars from around the globe.

Finally, we'd like to acknowledge the many hands behind the scenes that made this 59th meeting possible. Special thanks is due to Gabriel Iní Mendoza for his indispensable help with logistics, to the many anonymous reviewers who were invaluable in selecting the papers for presentation, and to the University of Chicago student volunteers who made the logistics of the conference come together. It is by your generous help and advice that the Chicago Linguistic Society can continue its tradition of scholarly excellence. We hope that this proceedings serves to inspire future scholars, future presenters, and future CLS organizers.

The CLS 59 Central Organizing Committee

Akshay Aitha

Kutay Serova

Madeline K. Snigaroff

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